

# Avi Vyas

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🔗 github.com/avi-vyas 🔗 avi-vyas.github.io/portfolio-application

## PROFESSIONAL EXPERIENCE

### Software Engineer (Backend Engineer)

01/2025 – Present | Noida, India

To The New (TTN) 🔗

- Developed and maintained scalable backend microservices using **Java** and **Spring Boot** for a platform serving **18M+ users**.
- Optimized **Elasticsearch-based search and autocomplete APIs**, reducing response time from **1s+ to ~80ms** while achieving **1,500+ TPS** with **~50% of production infrastructure**, significantly improving search performance, throughput, and infrastructure efficiency.
- **Currently developing an AI-powered Root Cause Analysis (RCA) microservice** that detects **KPI threshold breaches** and generates **evidence-based RCA** using application code, logs, and runtime data, with follow-up analysis capabilities, designed for adoption across **multiple teams and projects**.
- Regularly contribute to **production performance optimization and incident RCA**, investigating alerts, bottlenecks, and service-level issues across microservices and implementing targeted improvements to **reliability, latency, and system efficiency**.
- Developed an automated **free channel upgrade** solution enabling eligible subscribers to upgrade to **HD channels** at the same subscription price across multiple touchpoints, later generalized it into a **reusable microservice** adopted by multiple teams, eliminating duplicate implementations.
- Architected and developed **configurable promotions** across multiple implementation phases, supporting recurring and multiple promotion types through **business-rule-driven configuration**, reducing dependency on code changes for new promotional campaigns.

## SKILLS

**Languages** — Java (primary), Python (FastAPI)

**Frameworks & AI** — Spring Boot 4, Spring AI, Spring Security, Spring Data JPA, Hibernate, MCP , Spec Kit

**Databases & Messaging** — MySQL, PostgreSQL, Redis, Elasticsearch, RabbitMQ

**Cloud & DevOps** — AWS (EC2, S3), Docker, Git, GitLab, Maven, Gradle

**Testing & Core** — JUnit, Mockito, SLF4J, Microservices, REST APIs, Multithreading, OOP, SOLID, Data Structures

## PROJECTS

### E-Commerce Backend System 🔗

Java, Spring Boot, MySQL, Spring Security, JWT

- Developed a **role-based backend system** (Admin, Seller, Customer) with secure authentication and authorization using **Spring Security and JWT**.
- Designed and implemented **RESTful APIs** for product management, user management, and catalog services.
- Modeled complex product structures with **metadata-driven design**, enabling flexible and scalable catalog management.
- Implemented **pagination, validation, and exception handling** for robust API design.
- Followed **layered architecture and clean code principles** for maintainability and scalability.

### Doctor Appointment System 🔗

Python, FastAPI, MySQL, SQLAlchemy, JWT

- Built RESTful APIs for **appointment scheduling and booking systems** using FastAPI.
- Implemented **role-based access control (RBAC)** for doctors and patients with JWT authentication.
- Designed **time-slot scheduling system** with conflict detection and appointment lifecycle management.
- Used **ORM (SQLAlchemy)** for database interactions and modular architecture for maintainability.

## EDUCATION

**Bachelor of Technology (B.Tech) in Computer Science and Engineering**

2021 – 2025

BML Munjal University

CGPA: 8.14 / 10